WORKED SOLUTION

STEP II 2007, Question 3

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q3

The key idea

Everything rests on the standard integral of $1/(a^2+x^2)$, so the whole question is about recognising that shape after a substitution. In (i)(a) the numerator $\cos x$ is the derivative of $\sin x$, which is a strong hint. In (i)(b) the half-angle substitution turns $1+\sin^2 x$ into a quartic in t, and the point is to see that the same thing can be run backwards: in (ii) we spot which trigonometric integral produces the new quartic.

Solution

The standard integral

Let $a \neq 0$ and put $x = a\tan\theta$ with $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. As θ runs over this interval, x runs over every real number exactly once (whatever the sign of a), so the substitution is valid, and $\tan\theta = x/a$ gives $\theta = \arctan(x/a)$. Now $\mathrm{d}x = a\sec^2\theta\,\mathrm{d}\theta$ and

$$a^2 + x^2 = a^2(1+\tan^2\theta) = a^2\sec^2\theta,$$

so

$$\int \frac{1}{a^2+x^2}\,\mathrm{d}x = \int \frac{a\sec^2\theta}{a^2\sec^2\theta}\,\mathrm{d}\theta = \int \frac{1}{a}\,\mathrm{d}\theta = \frac{\theta}{a} + \text{constant} = \frac{1}{a}\arctan\frac{x}{a} + \text{constant},$$

as required.

Part (i)(a)

Substitute $u = \sin x$, so $\mathrm{d}u = \cos x\,\mathrm{d}x$, and u goes from 0 to 1 as x goes from 0 to $\frac{1}{2}\pi$. Using the standard integral with $a = 1$,

$$I = \int_0^1 \frac{1}{1+u^2}\,\mathrm{d}u = \Big[\arctan u\Big]_0^1 = \frac{\pi}{4}.$$

Part (i)(b)

With $t = \tan\frac{1}{2}x$ we have the standard half-angle results

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2}{1+t^2}\,\mathrm{d}t,$$

the last because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2\frac{1}{2}x = \frac{1}{2}(1+t^2)$. The limits $x = 0$ and $x = \frac{1}{2}\pi$ become $t = 0$ and $t = \tan\frac{1}{4}\pi = 1$. Next,

$$1 + \sin^2 x = \frac{(1+t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{1+6t^2+t^4}{(1+t^2)^2}.$$

Putting the pieces together, the factors of $1+t^2$ cancel completely:

$$I = \int_0^1 \frac{1-t^2}{1+t^2}\cdot\frac{(1+t^2)^2}{1+6t^2+t^4}\cdot\frac{2}{1+t^2}\,\mathrm{d}t = 2\int_0^1 \frac{1-t^2}{1+6t^2+t^4}\,\mathrm{d}t.$$

Dividing by 2 gives $\displaystyle\int_0^1 \frac{1-t^2}{1+6t^2+t^4}\,\mathrm{d}t = \tfrac{1}{2}I$, as required. (By (i)(a), its value is $\pi/8$.)

Part (ii)

? **Ask yourself** Where could the quartic $1+14t^2+t^4$ have come from?

In (i)(b) the $6t^2$ arose as $2t^2$ from squaring $1+t^2$, plus $4t^2 = (2t)^2$ from $\sin^2 x$. To get $14t^2$ we need an extra $12t^2 = 3(2t)^2$, that is, $3\sin^2 x$ in place of $\sin^2 x$. Indeed, with $t = \tan\frac{1}{2}x$,

$$1 + 3\sin^2 x = \frac{(1+t^2)^2 + 12t^2}{(1+t^2)^2} = \frac{1+14t^2+t^4}{(1+t^2)^2}.$$

So exactly the calculation of (i)(b), with $1+3\sin^2 x$ in place of $1+\sin^2 x$, gives

$$\int_0^{\frac{1}{2}\pi} \frac{\cos x}{1+3\sin^2 x}\,\mathrm{d}x = 2\int_0^1 \frac{1-t^2}{1+14t^2+t^4}\,\mathrm{d}t.$$

Evaluate the left-hand side as in (i)(a), with $u = \sin x$:

$$\int_0^1 \frac{1}{1+3u^2}\,\mathrm{d}u = \frac{1}{3}\int_0^1 \frac{1}{\frac{1}{3}+u^2}\,\mathrm{d}u = \frac{1}{3}\cdot\sqrt{3}\Big[\arctan\left(\sqrt{3}\,u\right)\Big]_0^1 = \frac{1}{\sqrt{3}}\cdot\frac{\pi}{3},$$

using the standard integral with $a = 1/\sqrt{3}$. Halving,

$$\int_0^1 \frac{1-t^2}{1+14t^2+t^4}\,\mathrm{d}t = \frac{\pi}{6\sqrt{3}} = \frac{\sqrt{3}\,\pi}{18}.$$

Checks

Simpson's rule gives $I \approx 0.785398$ (which is $\pi/4$), the integral in (i)(b) ≈ 0.392699 (which is $\pi/8$), and the integral in (ii) ≈ 0.302300, matching $\sqrt{3}\,\pi/18 \approx 0.302300$. A quick sanity check: both quartic integrands equal 1 at $t = 0$ and 0 at $t = 1$, and the one with $14t^2$ in the denominator is smaller throughout, so its integral should be smaller than $\pi/8 \approx 0.393$, as it is.

Common slips

- Forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}x$, which loses the factor of 2 in (i)(b).
- Writing $\int \frac{\mathrm{d}u}{1+3u^2} = \arctan(\sqrt{3}u)$ without the factor $\frac{1}{\sqrt{3}}$; take out the 3 first, or substitute $v = \sqrt{3}u$.
- Leaving the limits in x after substituting: $x = \frac{1}{2}\pi$ corresponds to $t = 1$.

Beyond STEP

There is a direct route too: dividing top and bottom by t^2 and putting $v = t + 1/t$ turns $\frac{1-t^2}{1+kt^2+t^4}$ into $-\frac{1}{v^2+k-2}$, another arctan integral. The general lesson, that rational functions of $\sin x$ and $\cos x$ always become rational functions of t under $t = \tan\frac{1}{2}x$ and so can always be integrated in principle, is the start of the systematic theory of integration in elementary terms. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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